

STUDENT NUMBER: _____

TEACHER'S NAME: _____

BAULKHAM HILLS HIGH SCHOOL

TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION

2007

MATHEMATICS

EXTENSION 1

*Time allowed – Two hours
(Plus five minutes reading time)*

GENERAL INSTRUCTIONS:

- Attempt **ALL** questions.
- Start each of the 7 questions on a new page.
- All necessary working should be shown.
- Write your student number at the top of each page of answer sheets.
- At the end of the exam, staple your answers in order behind the cover sheet.

QUESTION 1 (START ON A NEW PAGE)**Marks**

- (a) Find $\int \frac{dx}{\sqrt{9-25x^2}}$. **2**
- (b) Find the acute angle between the lines $y=3x+4$ and $2x+3y=6$. **2**
- (c) When the polynomial $P(x)=2x^3-ax+1$ is divided by $x+1$, the remainder is 2. Find a . **2**
- (d) Show that $\cot \theta - \cot 2\theta = \operatorname{cosec} 2\theta$ **3**
Hence find the exact value of $\cot 15^\circ$
- (e) Use the substitution $u = 2x + 1$ to find $\int x(2x+1)^{10} dx$ **3**

QUESTION 2 (START ON A NEW PAGE)

- (a) In the expansion of $\left(3x - \frac{1}{x^3}\right)^{12}$, find the term independent of x . **3**
- (b) (i) Sketch $y = 2\sin^{-1}(x-3)$, stating clearly the domain and range. **3**
(ii) Find the exact gradient of the function at the point where $x = 3.5$ **2**
- (c) A curve has parametric equations:
 $x = 3 \tan 2\theta$
 $y = \tan \theta$ **2**
Find its Cartesian equation.
- (d) Find $\int \cos^2 2x dx$. **2**

QUESTION 3 (START ON A NEW PAGE)**Marks**

(a) Simplify $\frac{6^x + 4^x}{3^x + 2^x}$ **2**

(b) Solve the equation $2x^3 - 17x^2 + 40x - 16 = 0$ given that it has a double root which is an integer. **4**

(c) Prove by mathematical induction that for all integers $n \geq 1$

$$2 \times 2^0 + 3 \times 2^1 + 4 \times 2^2 + \dots + (n+1) \times 2^{n-1} = n \times 2^n$$
3

(d) Find the range of values of x if the series

$$\frac{2}{1+3x} + \frac{6}{(1+3x)^2} + \frac{18}{(1+3x)^3} + \dots$$

has a limiting sum. **3**

QUESTION 4 (START ON A NEW PAGE)

(a) (i) Write $2\sqrt{3} \cos 2t - 2 \sin 2t$ in the form of $R \cos(2t + \alpha)$ **2**

(ii) A particle moves so that its displacement x metres is given by:

$$x = 2\sqrt{3} \cos 2t - 2 \sin 2t$$

Show that the motion is simple harmonic and state its period and amplitude **3**

(b) The equation $x + 2 \tan x = 0$ has a root near $\frac{3\pi}{4}$.

With $x = \frac{3\pi}{4}$ as a first approximation, find using Newton's method once, a second approximation to the root in terms of π **3**

(c) Show that $\frac{d}{d\theta} \left(\frac{1}{3} \tan^3 \theta \right) = \sec^4 \theta - \sec^2 \theta$ **2**

Hence or otherwise find the value of $\int_0^{\frac{\pi}{4}} \sec^4 \theta \, d\theta$ **2**

QUESTION 5 (START ON A NEW PAGE)**Marks****2**

(a) Find $\frac{d}{dx} \log_{10} (x^2 + 1)$

(b) The acceleration $\ddot{x} \text{ m/s}^2$ of a particle moving in a straight line is given by

$$\ddot{x} = 6(1 - x^2). \text{ Initially the particle is at } x = -3 \text{ and is moving with velocity}$$

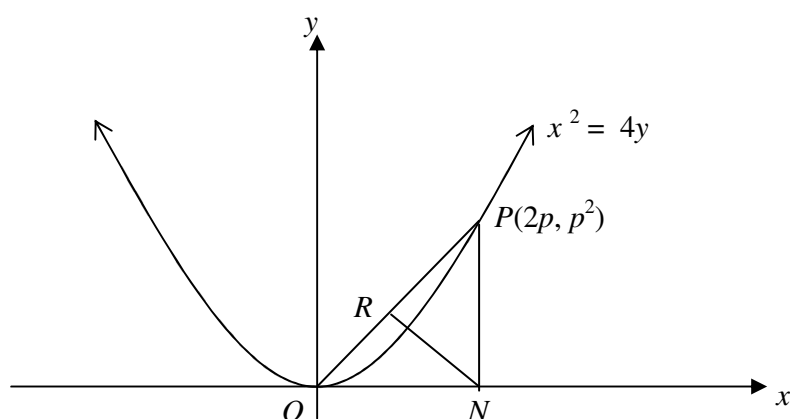
$$4 \text{ m/s. (i) By using the result } \ddot{x} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right) \text{ show that}$$

$$v^2 = 12x - 4x^3 - 56$$

3

(ii) Does the particle pass through the origin? Justify your answer.

(c)



$P(2p, p^2)$ is a variable point on the parabola $x^2 = 4y$. N is the foot of the perpendicular from P to the x axis. NR is perpendicular to OP

(i) Find the equation of OP

1

(ii) Find the equation of NR

1

(iii) Show that the R has coordinates $\left(\frac{8p}{p^2 + 4}, \frac{4p^2}{p^2 + 4} \right)$

2

(iv) Show that the locus of R is a circle and state its centre and radius

3**QUESTION 6 (START ON A NEW PAGE)**

(a) Solve the equation $e^x - e^{-x} = 1$ for x in the exact form.

3

(b) Given that $(1+ax)^7 + (1+bx)^7 = 2 + 21x + 609x^2 + \dots$
find the values of a and b

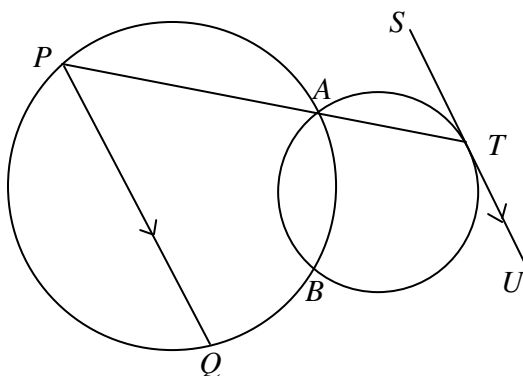
4

(c) Find the exact value of $\sin(2 \tan^{-1} \frac{3}{5})$ **2**

(d) Two circles intersect at A and B. STU is a tangent and is parallel to PQ.

Prove that the points Q, B, T are collinear.

3



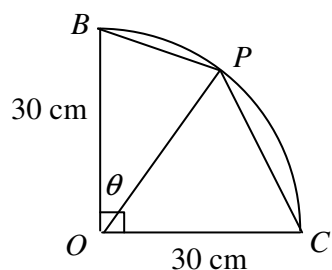
QUESTION 7 (START ON A NEW PAGE)

(a) If $f(x) = \ln(2x+3)$, find an expression for the inverse function $f^{-1}(x)$ 2

(b) P rotates about O along the arc BC at a constant rate of $\frac{\pi}{60}$ radians/minute.

(i) Show that the area of $OBPC$ is given by $A = 450(\sin \theta + \cos \theta)$ 2

(ii) Find the rate at which the area is changing when $\theta = \frac{\pi}{6}$ 2



(c) A projectile is fired from a point O on the ground with speed V m/s at an angle α . Given that the equations of motion are

$$x = V \cos \alpha t \quad \text{and} \quad y = V \sin \alpha t - \frac{gt^2}{2}$$

(i) Find the time of flight of the projectile. **1**

(ii) The projectile is climbing at an angle of 45° after a time T. Show that

$$T = \frac{V \sin \alpha - V \cos \alpha}{g} \quad 2$$

(iii) If T is $\frac{1}{3}$ of the time of flight, find α to the nearest degree. **3**

Ext 1 Trial 2007

Question 1

$$\begin{aligned} \text{a) } \int \frac{dx}{\sqrt{9-25x^2}} &= \int \frac{dx}{\sqrt{25(\frac{9}{25}-x^2)}} \\ &= \frac{1}{5} \sin^{-1} \frac{5x}{3} + C \end{aligned}$$

$$\begin{aligned} \text{(b) } m_1 &= 3 \quad m_2 = -\frac{2}{3} \\ \tan \theta &= \left| \frac{3 - (-\frac{2}{3})}{1 + 3(-\frac{2}{3})} \right| \\ &= \frac{11}{3} \\ \theta &= 74^\circ 45' \end{aligned}$$

$$\begin{aligned} \text{(c) } p(-1) &= 2 \\ 2(-1)^3 - a(-1) + 1 &= 2 \\ -2 + a + 1 &= 2 \\ a &= 3 \end{aligned}$$

$$\begin{aligned} \text{d) To prove } \cot \theta - \cot 2\theta &= \operatorname{cosec} 2\theta \\ \text{Proof: LHS} &= \frac{\cos \theta}{\sin \theta} - \frac{\cos 2\theta}{\sin 2\theta} \\ &= \frac{\sin 2\theta \cos \theta - \cos 2\theta \sin \theta}{\sin \theta \sin 2\theta} \\ &= \frac{\sin(2\theta - \theta)}{\sin \theta \sin 2\theta} \\ &= \frac{\sin \theta}{\sin \theta \sin 2\theta} \\ &= \operatorname{cosec} 2\theta = \text{RHS} \end{aligned}$$

$$\begin{aligned} \cot 15^\circ - \cot 30^\circ &= \operatorname{cosec} 30^\circ \\ \therefore \cot 15^\circ &= \cot 30^\circ + \operatorname{cosec} 30^\circ \\ &= \sqrt{3} + 2 \end{aligned}$$

$$\begin{aligned} \text{(e) } \int \frac{u-1}{2} \cdot u^{10} \cdot \frac{du}{2} & \quad u=2x+1 \\ & \quad du=2dx \\ &= \frac{1}{4} \int u^{11} - u^{10} du \quad x=\frac{u-1}{2} \\ &= \frac{1}{4} \left[\frac{u^{12}}{12} - \frac{u^{11}}{11} \right] + C \\ &= \frac{1}{4} \left[\frac{(2x+1)^{12}}{12} - \frac{(2x+1)^{11}}{11} \right] + C \end{aligned}$$

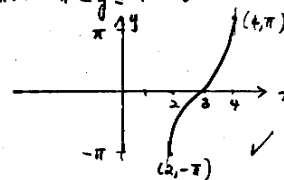
Question 2

$$\begin{aligned} \text{(a) } \left(3x - \frac{1}{x^3}\right)^{12} \\ T_{k+1} &= {}^{12}C_k (3x)^{12-k} \left(-\frac{1}{x^3}\right)^k \\ &= {}^{12}C_k 3^{12-k} (-1)^k x^{12-4k} \end{aligned}$$

For term indep of x $12-4k=0$
 $k=3$

$$\begin{aligned} \therefore T_4 &= {}^{12}C_3 3^9 (-1)^3 \\ &= -220(3)^9 \\ \text{or } -4330260 \end{aligned}$$

$$\begin{aligned} \text{(b) i) } -1 \leq x-3 \leq 1 \\ 2 \leq x \leq 4 \\ R: -\pi \leq y \leq \pi \end{aligned}$$



$$\begin{aligned} \text{ii) } y' &= \frac{2}{\sqrt{1-(x-3)^2}} \\ \text{When } x &= 3\frac{1}{2}, y' = \frac{2}{\sqrt{1-(\frac{1}{2})^2}} \\ &= \frac{2}{\sqrt{3}/2} = \frac{4}{\sqrt{3}} \text{ or } \frac{4\sqrt{3}}{3} \end{aligned}$$

$$\begin{aligned} \text{(c) } \frac{x}{3} &= \tan 2\theta, y = \tan \theta \\ \tan 2\theta &= \frac{2 \tan \theta}{1 - \tan^2 \theta} \\ \frac{x}{3} &= \frac{2y}{1-y^2} \\ x(1-y^2) &= 6y \end{aligned}$$

$$\begin{aligned} \text{(d) } \frac{1}{2} \int 1 + \cos 4x dx \\ &= \frac{1}{2} \left[x + \frac{\sin 4x}{4} \right] + C \end{aligned}$$

Q3

$$\begin{aligned} \text{(a) } \frac{6x^4 + x}{3x^2 + 2x} &= \frac{2x^3x + 2x^2x}{3x^2 + 2x} \\ &= \frac{2x^2(3x+2)}{3x^2 + 2x} \\ &= 2x \end{aligned}$$

$$\begin{aligned} \text{(b) } p(x) &= 2x^3 - 17x^2 + 40x - 16 \\ p'(x) &= 6x^2 - 34x + 40 \\ \text{For double root } p'(x) &= 0, p(x) = 0 \\ 2(3x^2 - 17x + 20) &= 0 \\ (3x-5)(x-4) &= 0 \\ x &= 4 \text{ or } 5/3 \\ \text{root is integer } \therefore x &= 4 \\ p(4) &= 0 \\ (x-4)^2(2x-1) &= 0 \\ \therefore x &= 4, 4, \frac{1}{2} \end{aligned}$$

$$\begin{aligned} \text{OR Let roots be } \alpha, \beta \\ 2\alpha + \beta &= \frac{17}{2} \\ \alpha^2 + 2\alpha\beta &= 20 \\ \alpha^2 + 2\alpha(\frac{17}{2} - 2\alpha) &= 20 \\ \alpha^2 + 17\alpha - 4\alpha^2 &= 20 \\ 3\alpha^2 - 17\alpha + 20 &= 0 \\ (3\alpha-5)(\alpha-4) &= 0 \\ \alpha \text{ integral } \therefore \alpha &= 4 \\ \therefore \text{Roots are } 4, 4, \frac{1}{2} \end{aligned}$$

$$\begin{aligned} \text{(c) To prove } 2x^2 + 3x^2 + \dots + (n+1)2 &= n \cdot 2^n \\ \text{Proof: test } n=1 \\ \text{LHS} &= 2 \times 2^0 = 2 \\ \text{RHS} &= 1 \cdot 2^1 = 2 \\ \therefore \text{true for } n=1 \end{aligned}$$

Assume true for $n=k$
ie $2x^2 + 3x^2 + \dots + (k+1)2 = k \cdot 2^k$
Prove true for $n=k+1$
ie $2x^2 + 3x^2 + \dots + (k+2)2 = (k+1)2^{k+1}$
 $= k \cdot 2^k + (k+2)2^k$
 $= 2^k(k+k+2)$
 $= 2^k(2k+2)$
 $= 2^k \cdot 2(k+1)$
 $= (k+1)2^{k+1}$
 $\therefore$ If true for $n=k$, it will be true for $n=k+1$
Since true for $n=1$, it will be true for $n=2, 3, \dots$ ie for all $n \geq 1$
(Lose 1 if conclusion not there)

$$\begin{aligned} \text{(d) } r &= \frac{3}{1+3x} \\ \text{For limiting sum } |r| &< 1 \\ \left| \frac{3}{1+3x} \right| &< 1 \\ 3 &< |1+3x| \\ 1+3x > 3 \quad \text{or} \quad -1-3x > 3 \\ x > \frac{2}{3} \quad \quad \quad x < -\frac{4}{3} \end{aligned}$$

Q4

$$\begin{aligned} \text{a) i) } 2\sqrt{3} \cos 2t - 2 \sin 2t \\ &= R \cos(2t + \alpha) \\ &= R \cos 2t \cos \alpha - R \sin 2t \sin \alpha \\ R \cos \alpha &= 2\sqrt{3} \\ R \sin \alpha &= 2 \\ \therefore R &= \sqrt{(2\sqrt{3})^2 + 2^2} = 4 \\ \tan \alpha &= \frac{1}{\sqrt{3}}, \quad \alpha = \frac{\pi}{6} \\ \therefore 2\sqrt{3} \cos 2t - 2 \sin 2t &= 4 \cos(2t + \frac{\pi}{6}) \end{aligned}$$

(ii) $x = 2\sqrt{3} \cos 2t - 2 \sin 2t$
 $= 4 \cos(2t + \frac{\pi}{2})$
 $\dot{x} = -8 \sin(2t + \frac{\pi}{2})$
 $\ddot{x} = -16 \cos(2t + \frac{\pi}{2})$
 $= -4x$
 of the form $-n^2x$
 $\therefore$ S.H.M
 period $= \frac{2\pi}{2} = \pi$
 amplitude $= 4$

(b) $\phi(x) = x + 2 \tan x$
 $\phi'(x) = 1 + 2 \sec^2 x$
 $x_1 = x_0 - \frac{\phi(x_0)}{\phi'(x_0)}$
 $= \frac{3\pi}{4} - \frac{\frac{3\pi}{4} + 2 \tan \frac{3\pi}{4}}{1 + 2 \sec^2 \frac{3\pi}{4}}$
 $= \frac{3\pi}{4} - \frac{\frac{3\pi}{4} - 2}{1 + 2(-\sqrt{2})^2}$
 $= \frac{3\pi}{4} - \frac{\frac{3\pi}{4} - 2}{5}$
 $= \frac{3\pi}{5} + \frac{2}{5}$

(c) $\frac{d}{dx} (\frac{1}{3} \tan^3 \theta)$
 $= \frac{1}{3} \cdot 3 \tan^2 \theta \sec^2 \theta$
 $= (\sec^2 \theta - 1) \sec^2 \theta$
 $= \sec^4 \theta - \sec^2 \theta$
 $\therefore \int_0^{\pi/4} \sec^4 \theta d\theta$
 $= [\frac{1}{3} \tan^3 \theta]_0^{\pi/4} + \int_0^{\pi/4} \sec^2 \theta d\theta$
 $= \frac{1}{3} \cdot 1^3 + [\tan \theta]_0^{\pi/4}$
 $= \frac{1}{3} + 1 - 0$
 $= \frac{4}{3}$

Q.5

(a) $\frac{d}{dx} \log_{10}(x^2+1)$
 $= \frac{1}{\ln 10} \cdot \frac{\ln(x^2+1)}{x^2+1}$
 $= \frac{1}{\ln 10} \cdot \frac{2x}{x^2+1}$

(b) $\ddot{x} = 6(1-x^2)$
 $\frac{d}{dx} (\frac{1}{2} v^2) = 6 - 6x^2$
 $\frac{v^2}{2} = 6x - 2x^3 + C$
 $x = -3, v = 4 \therefore 8 = -18 + 54 + C$
 $-28 = C$
 $\therefore v^2 = 12x - 4x^3 - 56$

(ii) If $x=0, v^2 = -56$ not possible
 $\therefore$ No

(c) (i) OP: $y = \frac{p}{2p} x \therefore y = \frac{p}{2} x$

(ii) $y - 0 = -\frac{2}{p}(x - 2p)$
 $y = -\frac{2}{p}x + 4$

(iii) Sub (ii) into (i)
 $\frac{p}{2}x = -\frac{2}{p}x + 4$
 $p^2x = -4x + 8p$
 $x(p^2 + 4) = 8p$
 $x = \frac{8p}{p^2 + 4}$
 $y = \frac{p}{2} \cdot \frac{8p}{p^2 + 4} = \frac{4p^2}{p^2 + 4}$

(iv) $\frac{x}{y} = \frac{2}{p} \therefore p = \frac{2y}{x}$
 Sub in x $x = \frac{8(\frac{2y}{x})}{\frac{4y^2}{x^2} + 4} = \frac{16y}{4y^2 + 4x^2}$
 locus is $4x^2 + 4y^2 = 16y$
 $x^2 + (y-2)^2 = 4$
 circle $(0, 2), r = 2$

Q.6

(a) $e^x - \frac{1}{e^x} = 1$
 $e^{2x} - e^x - 1 = 0$
 $u = e^x$
 $u^2 - u - 1 = 0$
 $e^x = u = \frac{1 \pm \sqrt{1+4(1)(-1)}}{2}$
 $e^x > 0 \therefore e^x = \frac{1+\sqrt{5}}{2}$
 $x = \ln(\frac{1+\sqrt{5}}{2})$

(b) $(1+ax)^7 + (1+bx)^7$
 $= 1 + 7C_1ax + 7C_2a^2x^2 + \dots$
 $+ 1 + 7C_1bx + 7C_2b^2x^2 + \dots$

$\therefore 7a+7b = 21$ or $a+b=3$
 $7C_2a^2+7C_2b^2 = 609$ or $a^2+b^2=29$

$a=3-b$
 $9-6b+b^2+b^2=29$
 $2b^2-6b-20=0$
 $b^2-3b-10=0$
 $(b-5)(b+2)=0$
 $\therefore b=5$ or -2 any pair
 $a=-2$ or 5

(c) $\sin(2 + \tan^{-1} \frac{3}{5})$
 $= 2 \sin \alpha \cos \alpha$
 $= 2(-\frac{3}{5}) \cdot \frac{5}{\sqrt{34}} = -\frac{30}{34} = -\frac{15}{17}$

(d) Join AB, QB, BT
 Let $\angle APQ = \alpha$
 $\angle ABQ = 180^\circ - \alpha$ (opp. $\angle$ s of cyclic quad. supp.)
 $\angle STA = \angle APQ = \alpha$ (alt. $\angle$ s on ll lines)
 $\angle ABT = \angle STA = \alpha$ ($\angle$ between tangent and chord = $\angle$ in alt. seg.)

$\therefore \angle ABQ + \angle ABT$
 $= 180^\circ - \alpha + \alpha$
 $= 180^\circ$
 $\therefore$ Q, B, T are collinear

Q.7 (a) Let $x = \ln(2y+3)$
 $e^x = 2y+3$
 $\phi'(x) = y = \frac{e^x-3}{2}$

(b) (i) Area of $\triangle BPC$
 $=$ area of $\triangle OPB +$ area of $\triangle OPC$
 $= \frac{1}{2} \cdot 30^2 \sin \theta + \frac{1}{2} \cdot 30^2 \sin(90^\circ - \theta)$
 $= 450(\sin \theta + \cos \theta)$

(ii) $\frac{dA}{d\theta} = 450(\cos \theta - \sin \theta)$
 $\frac{d\theta}{dt} = \frac{\pi}{60}$ rad/min

$\therefore \frac{dA}{dt} = 450(\cos \frac{\pi}{6} - \sin \frac{\pi}{6}) \times \frac{\pi}{60} \text{ cm}^2/\text{min}$
 $= 450(\frac{\sqrt{3}}{2} - \frac{1}{2}) \times \frac{\pi}{60} \text{ cm}^2/\text{min}$
 $= \frac{15\pi}{4}(\sqrt{3}-1)$ or $8.624 \text{ cm}^2/\text{min}$
 either one

(c) (i) Time of flight when $y=0$
 $t(V \sin \alpha - g \frac{t}{2}) = 0$
 $\therefore t = \frac{2V \sin \alpha}{g}$

(ii) $\tan 45^\circ = \frac{h}{x}$
 $1 = \frac{V \sin \alpha - gT}{V \cos \alpha}$
 $V \cos \alpha = V \sin \alpha - gT$
 $\therefore T = \frac{V \sin \alpha - V \cos \alpha}{g}$

(iii) $\frac{V \sin \alpha - V \cos \alpha}{g} = \frac{1}{3} \cdot \frac{2V \sin \alpha}{g}$
 $3 \sin \alpha - 3 \cos \alpha = 2 \sin \alpha$
 $\tan \alpha = 3$
 $\alpha = 72^\circ$